



Routing Protocols Guide



From Fundamentals to Advanced Concepts



👤 Prepared by: Imteyaz Ali

📅 April 15, 2025

Network Engineering Series



1 Networking Fundamentals

Understanding networking fundamentals is crucial for mastering routing protocols. This section breaks down the core concepts that underpin modern networks.

1.1 OSI Model and TCP/IP Model

The OSI and TCP/IP models are frameworks that define how data flows across networks.

1.1.1 Why These Models Matter

- ****Standardization****: Ensures interoperability between devices from different vendors.
- ****Troubleshooting****: Helps isolate issues to specific layers (e.g., physical vs. application layer problems).

1.1.2 OSI Model Layers

- **Layer 1 (Physical)**: Transmits raw bits over cables, switches, or wireless signals. *Example*: Ethernet cables and Wi-Fi signals operate here.
- **Layer 2 (Data Link)**: Uses MAC addresses to forward frames. *Example*: Switches use MAC tables to direct traffic within a LAN.
- **Layer 3 (Network)**: Handles IP addressing and routing. *Example*: Routers use IP addresses to forward packets between networks.
- **Layer 4 (Transport)**: Manages end-to-end communication (TCP/UDP). *Why It Matters*: TCP ensures reliable delivery (e.g., file transfers), while UDP prioritizes speed (e.g., video streaming).

1.1.3 TCP/IP Model

Simplifies the OSI model into four layers:

- **Network Interface**: Combines OSI Layers 1 and 2.
- **Internet**: Equivalent to OSI Layer 3.
- **Transport**: Equivalent to OSI Layer 4.
- **Application**: Combines OSI Layers 5–7 (e.g., HTTP, DNS).

1.2 IP Addressing

IP addresses uniquely identify devices on a network.

1.2.1 IPv4 vs. IPv6

- **IPv4**: - 32-bit addresses (e.g., 192.168.1.1). - Limited to 4.3 billion addresses. *Why It's a Problem*: IPv4 exhaustion led to the adoption of IPv6.
- **IPv6**: - 128-bit addresses (e.g., 2001:0db8::1). - Solves address exhaustion with 340 undecillion addresses.



1.2.2 Subnetting

Divides a network into smaller subnets to reduce broadcast domains and improve efficiency.

Example: Split 192.168.1.0/24 into two subnets:

- Subnet 1: 192.168.1.0/25 (126 hosts).
- Subnet 2: 192.168.1.128/25 (126 hosts).

Why Subnetting Matters: - Reduces network congestion. - Enhances security by isolating sensitive devices.

1.2.3 CIDR Notation

Shorthand for subnet masks. Example:

- /24 = 255.255.255.0 (254 usable hosts).
- /16 = 255.255.0.0 (65,534 usable hosts).

1.3 Routers and Routing Tables

Routers are the backbone of inter-network communication. They use routing tables to make forwarding decisions.

1.3.1 Routing Table Components

- **Destination Network:** The network prefix (e.g., 192.168.1.0/24).
- **Next Hop:** The IP address of the next router or interface.
- **Metric:** A value indicating the "cost" of the route (e.g., hop count).
- **Administrative Distance:** Trustworthiness of the route source (lower is better).

1.3.2 Example Routing Table

1	Destination	Gateway	Genmask	Flags	Metric
	Iface				
2	192.168.1.0	*	255.255.255.0	U	0
	eth0				
3	default	192.168.1.1	0.0.0.0	UG	100
	eth0				

Why Administrative Distance Matters: - A static route (AD 1) is preferred over OSPF (AD 110) because it's manually configured and trusted more.



2 Static and Dynamic Routing

Routing determines how data travels across networks. This section explores static and dynamic routing, their use cases, and their differences.

2.1 Static Routing

Static routes are manually configured by administrators. They are simple but lack scalability.

2.1.1 Configuration Example

```
1 ! Configure a static route for 192.168.2.0/24 via 192.168.1.2
2 Router(config)# ip route 192.168.2.0 255.255.255.0 192.168.1.2
3
4 ! Configure a default route (gateway of last resort)
5 Router(config)# ip route 0.0.0.0 0.0.0.0 192.168.1.1
```

2.1.2 Why Use Static Routing?

- **Predictability**: Routes never change unless manually updated.
- **Security**: No risk of unauthorized route advertisements.
- **Use Cases**:
 - Small offices with fixed network layouts.
 - Backup routes for critical connections.

2.1.3 Limitations

- **Manual Maintenance**: A single topology change requires manual updates.
- **No Fault Tolerance**: If the next-hop fails, traffic is dropped.

2.2 Dynamic Routing

Dynamic routing protocols automatically discover and adapt to network changes.

2.2.1 Key Concepts

- Convergence**:
 - The time taken for routers to agree on the best path after a change.
 - *Example*: OSPF converges in seconds, while RIP may take minutes.
- Administrative Distance (AD)**:
 - Determines the trustworthiness of a route source.
 - **Lower AD = Higher Priority**.
 - *Example AD Values*:
 - Connected Interface: 0
 - OSPF: 110
 - RIP: 120
- Metrics**:
 - Used to compare routes from the same protocol.
 - *Examples*:
 - RIP: Hop count.
 - OSPF: Bandwidth-based cost.



2.2.2 Why Dynamic Routing Matters

- **Scalability**: Adapts to large, complex networks. - **Fault Tolerance**: Automatically reroutes traffic if a link fails.

2.3 Dynamic Routing Protocols: Types

Dynamic protocols are categorized into **IGPs** (Interior Gateway Protocols) and **EGPs** (Exterior Gateway Protocols).

2.3.1 Interior Gateway Protocols (IGPs)

Used within a single autonomous system (AS): - **Distance Vector Protocols**: - Share entire routing tables periodically. - *Example*: RIP. - **Link-State Protocols**: - Share topology information to build a network map. - *Example*: OSPF. - **Hybrid Protocols**: - Combine features of both. - *Example*: EIGRP.

2.3.2 Exterior Gateway Protocols (EGPs)

Used between autonomous systems (e.g., ISPs): - **Border Gateway Protocol (BGP)**: - The de facto standard for internet routing. - Uses path attributes (e.g., *AS_PATH*) to select routes.

2.4 Static vs. Dynamic Routing: Comparison

Feature	Static Routing	Dynamic Routing
Configuration	Manual	Automatic
Scalability	Poor	Excellent
Fault Tolerance	None	Built-in
Resource Usage	Low (no updates)	High (periodic updates)
Use Case	Small networks, back-ups	Large/complex networks

2.4.1 Real-World Example

- **Static Routing**: A branch office with a single internet connection. - **Dynamic Routing**: A data center with multiple redundant links.



3 Interior Gateway Protocols (IGPs)

IGPs manage routing within a single autonomous system (AS). They are divided into three categories: distance vector, link-state, and hybrid protocols.

3.1 Distance Vector Protocols

Share routing updates periodically and use hop count or similar metrics.

3.1.1 RIP (Routing Information Protocol)

Characteristics: - Metric: Hop count (max 15 hops). - Updates: Broadcast every 30 seconds. - *Why It Matters:* Simple to configure but unsuitable for large networks.

Configuration Example:

```
1 ! Enable RIP and advertise networks
2 Router(config)# router rip
3 Router(config-router)# version 2
4 Router(config-router)# network 192.168.1.0
5 Router(config-router)# network 192.168.2.0
```

Limitations: - Slow convergence (prone to routing loops). - No support for VLSM (RIP v1) or authentication (RIP v1).

3.2 Link-State Protocols

Build a complete network topology and use algorithms like Dijkstra's SPF.

3.2.1 OSPF (Open Shortest Path First)

Characteristics: - Metric: Cost (bandwidth-based). - Hierarchical design with areas (Area 0 is the backbone). - *Why It Matters:* Scalable and fast-converging for large networks.

Configuration Example:

```
1 ! Configure OSPF with Area 0
2 Router(config)# router ospf 1
3 Router(config-router)# network 192.168.1.0 0.0.0.255 area 0
4 Router(config-router)# network 192.168.2.0 0.0.0.255 area 1
```

Key Features: - ****LSAs (Link-State Advertisements)**:** Flood topology changes. - ****DR/BDR Election**:** Reduces OSPF traffic in broadcast networks.

3.2.2 IS-IS (Intermediate System to Intermediate System)

Characteristics: - Similar to OSPF but uses CLNS (Connectionless Network Service). - *Why It Matters:* Preferred in service provider networks for scalability.



3.3 Hybrid Protocols

Combine features of distance vector and link-state protocols.

3.3.1 EIGRP (Enhanced Interior Gateway Routing Protocol)

Characteristics: - Metric: Bandwidth, delay, reliability, load. - Fast convergence via DUAL (Diffusing Update Algorithm). - *Why It Matters:* Efficient for Cisco-centric networks.

Configuration Example:

```
1 ! Enable EIGRP and advertise networks
2 Router(config)# router eigrp 100
3 Router(config-router)# network 192.168.1.0
4 Router(config-router)# network 192.168.2.0
```

Advanced Features: - ****Unequal-Cost Load Balancing**:** Uses variance command. - ****Neighbor Discovery**:** Hello packets every 5 seconds.

3.4 IGP Comparison Table

Protocol	Type	Metric	Convergence
RIP	Distance Vector	Hop count	Slow
OSPF	Link-State	Bandwidth	Fast
EIGRP	Hybrid	Composite	Very Fast
IS-IS	Link-State	Default (bandwidth)	Fast

3.4.1 Real-World Example: OSPF in a Multi-Area Network

- **Scenario:** A company with three locations (HQ, Branch 1, Branch 2).
- **Design:** - HQ uses Area 0 (backbone). - Branches use Areas 1 and 2.
- **Benefit:** - Localized updates (Area 1 changes don't affect Area 2). - Reduced SPF calculations.



4 Exterior Gateway Protocols (EGPs)

EGPs manage routing **between autonomous systems (AS)**, such as ISPs or large organizations. The most widely used EGP is **BGP**.

4.1 Autonomous Systems (AS)

- **Definition**: A collection of IP networks under the control of a single organization. - **AS Numbers (ASN)**: - Public ASNs: Assigned by IANA (e.g., 'AS65001'). - Private ASNs: Used internally (e.g., 'AS64512'–'AS65535').

4.1.1 Why ASNs Matter

- **Global Routing**: ASNs uniquely identify networks on the internet. - **Policy Enforcement**: ISPs use ASNs to control traffic flow (e.g., preferring one peer over another).

4.2 BGP (Border Gateway Protocol)

BGP is the de facto protocol for internet routing. It uses **path-vector logic** to select routes based on policies.

4.2.1 Key Concepts

1. **Peering**: - **eBGP**: Between different ASNs (e.g., ISP to ISP). - **iBGP**: Within the same ASN (e.g., between routers in a data center). 2. **Path Attributes**: - **AS_PATH**: List of ASNs a route has traversed. - **NEXT_HOP**: IP address of the next router. - **LOCAL_PREF**: Preferred routes within an AS. - **MED (Multi-Exit Discriminator)**: Influences inbound traffic.

4.2.2 Configuration Example

```

1 ! Configure BGP for AS 65001
2 Router(config)# router bgp 65001
3 Router(config-router)# neighbor 203.0.113.2 remote-as 65002
4 Router(config-router)# network 192.168.1.0 mask 255.255.255.0
5
6 ! Advertise a prefix with a community
7 Router(config-router)# route-map SET_COMMUNITY permit 10
8 Router(config-route-map)# set community 65001:100
9 Router(config-router)# neighbor 203.0.113.2 route-map
   SET_COMMUNITY out

```

4.2.3 Why BGP Matters

- **Internet Backbone**: BGP powers the global routing table (800,000 prefixes). - **Policy Control**: Organizations can enforce traffic policies (e.g., "use ISP A for 80% of traffic").



4.3 BGP vs. IGP: Key Differences

Feature	BGP (EGP)	OSPF/EIGRP (IGP)
Scope	Between ASNs	Within an AS
Metric	Path attributes (policy-driven)	Bandwidth/hop count
Convergence	Slow (minutes)	Fast (seconds)
Use Case	Internet, large-scale networks	Enterprise/data center networks

4.4 Real-World Example: BGP in Action

****Scenario****: An organization connects to two ISPs for redundancy. ****BGP Configuration****: 1. Assign a public ASN (e.g., ‘AS65001’). 2. Advertise the organization’s public IP range (e.g., ‘203.0.113.0/24’). 3. Use ****LOCAL_PREF *toprioritizeoneISPovertheother****.

****Result****: - Traffic exits via the preferred ISP unless it fails. - The internet learns the organization’s routes via BGP.

4.5 Advanced BGP Features

1. ****Route Aggregation****: - Combine multiple prefixes into a single advertisement. “bash Router(config-router) aggregate-address 203.0.113.0 255.255.255.0



5 Advanced Routing Concepts

This section explores advanced techniques and protocols that build on IGPs and EGPs.

5.1 Route Redistribution

Integrates routes from different protocols (e.g., OSPF into EIGRP).

5.1.1 Why It Matters

- Enables communication between networks using different routing protocols. - Common in multi-protocol environments (e.g., merging legacy and modern networks).

5.1.2 Configuration Example

```
1 ! Redistribute OSPF routes into EIGRP
2 Router(config)# router eigrp 100
3 Router(config-router)# redistribute ospf 1 metric 10000 100 255
   1 1500
4
5 ! Redistribute static routes into OSPF
6 Router(config)# router ospf 1
7 Router(config-router)# redistribute static subnets
```

5.1.3 Challenges

- **Routing Loops**: Mitigate with route filters or tags. - **Metric Mismatch**: Use 'default-metric' to standardize values.

5.2 MPLS (Multiprotocol Label Switching)

MPLS improves packet forwarding by adding labels to traffic.

5.2.1 Key Concepts

- **Labels**: Short, fixed-length identifiers swapped at each hop. - **LSP (Label Switched Path)**: Predefined path through the network. - **Why It Matters**: Enables traffic engineering and VPN services.

5.2.2 Configuration Example

```
1 ! Enable MPLS on an interface
2 Router(config)# interface GigabitEthernet0/1
3 Router(config-if)# mpls ip
4
```



```
5 ! Configure LDP (Label Distribution Protocol)
6 Router(config)# mpls ldp router-id Loopback0
```

5.3 Multicast Routing (PIM)

Enables efficient distribution of traffic to multiple hosts.

5.3.1 PIM Modes

- **PIM-DM (Dense Mode)**: Floods traffic and prunes unnecessary paths. - **PIM-SM (Sparse Mode)**: Uses rendezvous points (RPs) for efficient forwarding.

5.3.2 Configuration Example

```
1 ! Enable PIM-SM on interfaces
2 Router(config)# interface GigabitEthernet0/0
3 Router(config-if)# ip pim sparse-mode
4
5 ! Configure a static RP
6 Router(config)# ip pim rp-address 192.168.1.100
```

5.4 Policy-Based Routing (PBR)

Overrides routing decisions based on policies (e.g., source IP).

5.4.1 Configuration Example

```
1 ! Route traffic from 192.168.1.0/24 via 10.0.0.1
2 Router(config)# access-list 10 permit 192.168.1.0 0.0.0.255
3 Router(config)# route-map PBR_MAP permit 10
4 Router(config-route-map)# match ip address 10
5 Router(config-route-map)# set ip next-hop 10.0.0.1
6 Router(config)# interface GigabitEthernet0/0
7 Router(config-if)# ip policy route-map PBR_MAP
```

5.5 Security in Routing

Protect routing protocols from attacks.

5.5.1 Techniques

- **Route Filtering**: Use prefix lists or ACLs to block invalid routes. - **Authentication**: Secure OSPF/EIGRP with MD5 or SHA.



5.5.2 Example: OSPF Authentication

```
1 ! Enable MD5 authentication on OSPF
2 Router(config)# interface GigabitEthernet0/0
3 Router(config-if)# ip ospf authentication message-digest
4 Router(config-if)# ip ospf message-digest-key 1 md5
   SECRET_PASSWORD
```

5.6 IPv6 Routing

IPv6 introduces new routing considerations.

5.6.1 Configuration Example

```
1 ! Enable OSPFv3 for IPv6
2 Router(config)# ipv6 router ospf 1
3 Router(config-rtr)# router-id 1.1.1.1
4 Router(config)# interface GigabitEthernet0/0
5 Router(config-if)# ipv6 ospf 1 area 0
```

5.7 Real-World Example: MPLS in a Service Provider Network

Scenario: A service provider uses MPLS to offer VPN services to customers. **Design:**

- Core Routers:** Use MPLS to forward traffic based on labels.
- PE Routers:** Assign labels to customer traffic (e.g., Customer A: Label 20).
- CE Routers:** Exchange routes with the provider via BGP or OSPF.

Benefit:

- Customers' traffic is isolated and prioritized.
- Scalable for thousands of clients.

5.7.1 Why Advanced Concepts Matter

- **Traffic Engineering:** MPLS optimizes bandwidth usage.
- **Security:** Route filtering and authentication prevent attacks.
- **Flexibility:** PBR and redistribution enable customized routing policies.

